## Description

### Image





### Caption

1. Close-up of the material's surface. © Chris Lefteri 2. Bike seats with polyurethane cores. © Chris Lefteri

### The material

Think of polyurethanes and you think of the soft, the stretchy, materials and fabrics (Lycra or Spandex). Like PVC, polyurethanes have thermoplastic, elastomeric and thermosetting grades. They are easily foamed; some 40% of all PU is made into foam by mixing it with a blowing agent. The foams can be open- or closed-cell, microcellular or filter grades. They are the strongest of elastomers.

### Compositional summary

(CO-NH-R-NH-CO-O-R-O)n

## General properties

| | | | | | |
|---|---|---|---|---|---|
| Density | | 1.19e3 | - | 1.21e3 | kg/m^3 |
| Price | * | 1.31 | - | 1.67 | GBP/kg |
| Date first used | | 1941 | | | |

## Mechanical properties

| | | | | | |
|---|---|---|---|---|---|
| Young's modulus | | 0.0025 | - | 0.03 | GPa |
| Shear modulus | * | 8.33e-4 | - | 0.01 | GPa |
| Bulk modulus | * | 1.5 | - | 2 | GPa |
| Poisson's ratio | * | 0.47 | - | 0.49 | |
| Yield strength (elastic limit) | | 40 | - | 51 | MPa |
| Tensile strength | | 40 | - | 51 | MPa |
| Compressive strength | * | 48 | - | 61.2 | MPa |
| Elongation | | 500 | - | 750 | % strain |
| Hardness - Vickers | * | 12 | - | 14 | HV |

Values marked * are estimates.
ANSYS, Inc. provides no warranty for this data.

| | | | | | |
|---|---|---|---|---|---|
| Fatigue strength at 10^7 cycles | * | 16 | - | 20.4 | MPa |
| Fracture toughness | | 0.192 | - | 1.29 | MPa.m^0.5 |
| Mechanical loss coefficient (tan delta) | | 0.05 | - | 0.1 | |

## Thermal properties

| | | | | | |
|---|---|---|---|---|---|
| Glass temperature | * | -73.2 | - | -23.2 | °C |
| Maximum service temperature | | 79.9 | - | 105 | °C |
| Minimum service temperature | | -65.2 | - | -30.2 | °C |
| Thermal conductor or insulator? | | Good insulator | | | |
| Thermal conductivity | | 0.28 | - | 0.3 | W/m.°C |
| Specific heat capacity | | 1.65e3 | - | 1.7e3 | J/kg.°C |
| Thermal expansion coefficient | | 160 | - | 165 | µstrain/°C |

## Electrical properties

| | | | | | |
|---|---|---|---|---|---|
| Electrical conductor or insulator? | | Good insulator | | | |
| Electrical resistivity | | 1e17 | - | 3.16e18 | µohm.cm |
| Dielectric constant (relative permittivity) | | 5 | - | 8 | |
| Dissipation factor (dielectric loss tangent) | | 0.02 | - | 0.09 | |
| Dielectric strength (dielectric breakdown) | | 14 | - | 20 | MV/m |

## Optical properties

| | | | | | |
|---|---|---|---|---|---|
| Transparency | | Transparent | | | |
| Refractive index | * | 1.48 | - | 1.5 | |

## Critical Materials Risk

| | | | | | |
|---|---|---|---|---|---|
| High critical material risk? | | No | | | |

## Processability

| | | | | | |
|---|---|---|---|---|---|
| Castability | | 4 | - | 5 | |
| Moldability | | 4 | - | 5 | |
| Machinability | | 2 | - | 3 | |
| Weldability | | 1 | | | |

## Durability: water and aqueous solutions

| | | | | | |
|---|---|---|---|---|---|
| Water (fresh) | | Excellent | | | |
| Water (salt) | | Acceptable | | | |
| Soils, acidic (peat) | | Unacceptable | | | |
| Soils, alkaline (clay) | | Limited use | | | |
| Wine | | Limited use | | | |

## Durability: acids

| | |
|---|---|
| Acetic acid (10%) | Unacceptable |
| Acetic acid (glacial) | Unacceptable |
| Citric acid (10%) | Excellent |
| Hydrochloric acid (10%) | Limited use |
| Hydrochloric acid (36%) | Unacceptable |
| Hydrofluoric acid (40%) | Unacceptable |
| Nitric acid (10%) | Limited use |
| Nitric acid (70%) | Unacceptable |
| Phosphoric acid (10%) | Limited use |
| Phosphoric acid (85%) | Unacceptable |
| Sulfuric acid (10%) | Limited use |
| Sulfuric acid (70%) | Unacceptable |

## Durability: alkalis

| | |
|---|---|
| Sodium hydroxide (10%) | Limited use |
| Sodium hydroxide (60%) | Unacceptable |

## Durability: fuels, oils and solvents

| | |
|---|---|
| Amyl acetate | Unacceptable |
| Benzene | Unacceptable |
| Carbon tetrachloride | Unacceptable |
| Chloroform | Unacceptable |
| Crude oil | Limited use |
| Diesel oil | Limited use |
| Lubricating oil | Limited use |
| Paraffin oil (kerosene) | Excellent |
| Petrol (gasoline) | Limited use |
| Silicone fluids | Excellent |
| Toluene | Limited use |
| Turpentine | Unacceptable |
| Vegetable oils (general) | Excellent |
| White spirit | Unacceptable |

## Durability: alcohols, aldehydes, ketones

| | |
|---|---|
| Acetaldehyde | Unacceptable |
| Acetone | Unacceptable |
| Ethyl alcohol (ethanol) | Unacceptable |
| Ethylene glycol | |

| | |
|---|---|
| | Unacceptable |
| Formaldehyde (40%) | Unacceptable |
| Glycerol | Excellent |
| Methyl alcohol (methanol) | Unacceptable |

## Durability: halogens and gases

| | |
|---|---|
| Chlorine gas (dry) | Unacceptable |
| Fluorine (gas) | Unacceptable |
| O2 (oxygen gas) | Unacceptable |
| Sulfur dioxide (gas) | Unacceptable |

## Durability: built environments

| | |
|---|---|
| Industrial atmosphere | Excellent |
| Rural atmosphere | Excellent |
| Marine atmosphere | Excellent |
| UV radiation (sunlight) | Poor |

## Durability: flammability

| | |
|---|---|
| Flammability | Slow-burning |

## Durability: thermal environments

| | |
|---|---|
| Tolerance to cryogenic temperatures | Unacceptable |
| Tolerance up to 150 C (302 F) | Acceptable |
| Tolerance up to 250 C (482 F) | Unacceptable |
| Tolerance up to 450 C (842 F) | Unacceptable |
| Tolerance up to 850 C (1562 F) | Unacceptable |
| Tolerance above 850 C (1562 F) | Unacceptable |

## Primary material production: energy, climate change and water

| | | | | |
|---|---|---|---|---|
| Climate change (CO2-eq) (virgin grade production) | * 3.05 | - | 3.37 | kg/kg |
| Embodied energy (virgin grade production) | * 70.2 | - | 77.4 | MJ/kg |
| Water usage | * 93.5 | - | 103 | l/kg |

## Material processing: energy

| | | | | |
|---|---|---|---|---|
| Polymer molding (injection, blow), embodied energy | * 16.1 | - | 17.7 | MJ/kg |
| Machining, grinding, embodied energy | * 26.4 | - | 32.4 | MJ/kg |

## Material processing: climate change

| | | | | |
|---|---|---|---|---|
| Polymer molding (injection, blow), climate change (CO2-eq) | * 1.05 | - | 1.16 | kg/kg |
| Machining, grinding, climate change (CO2-eq) | * 1.73 | - | 2.12 | kg/kg |

## Recycling and end of life

| | | | | |
|---|---|---|---|---|
| Recycle | ✗ | | | |
| Functional recycle | ✗ | | | |
| Combust for energy recovery | ✓ | | | |
| Heat of combustion (net) | * 21.8 | - | 22.9 | MJ/kg |
| Combustion CO2 | * 2 | - | 2.1 | kg/kg |
| Landfill | ✓ | | | |
| Biodegrade | ✗ | | | |
| Toxicity rating | Non-toxic | | | |
| A renewable resource? | ✗ | | | |

### Environmental notes

Polyurethane elastomers are thermosets, and thus cannot be recycled. Their disposal creates an environmental problem.

### Recycle mark




## Supporting information

### Design guidelines

Urethanes have exceptional strength (up to 48 MPa) and abrasion resistance, low compression set and good fuel resistance. They have useful properties from -55 C to 90 C

### Technical notes

Urethane elastomers (elPU) are co-polymers of diisocyanate and polyester.

### Typical uses

Cushioning, packaging, shoe soles, tires, fuel hoses, gears, bearings, car bumpers, adhesives, fabric-coating.

## Links

ProcessUniverse

Producers

Reference